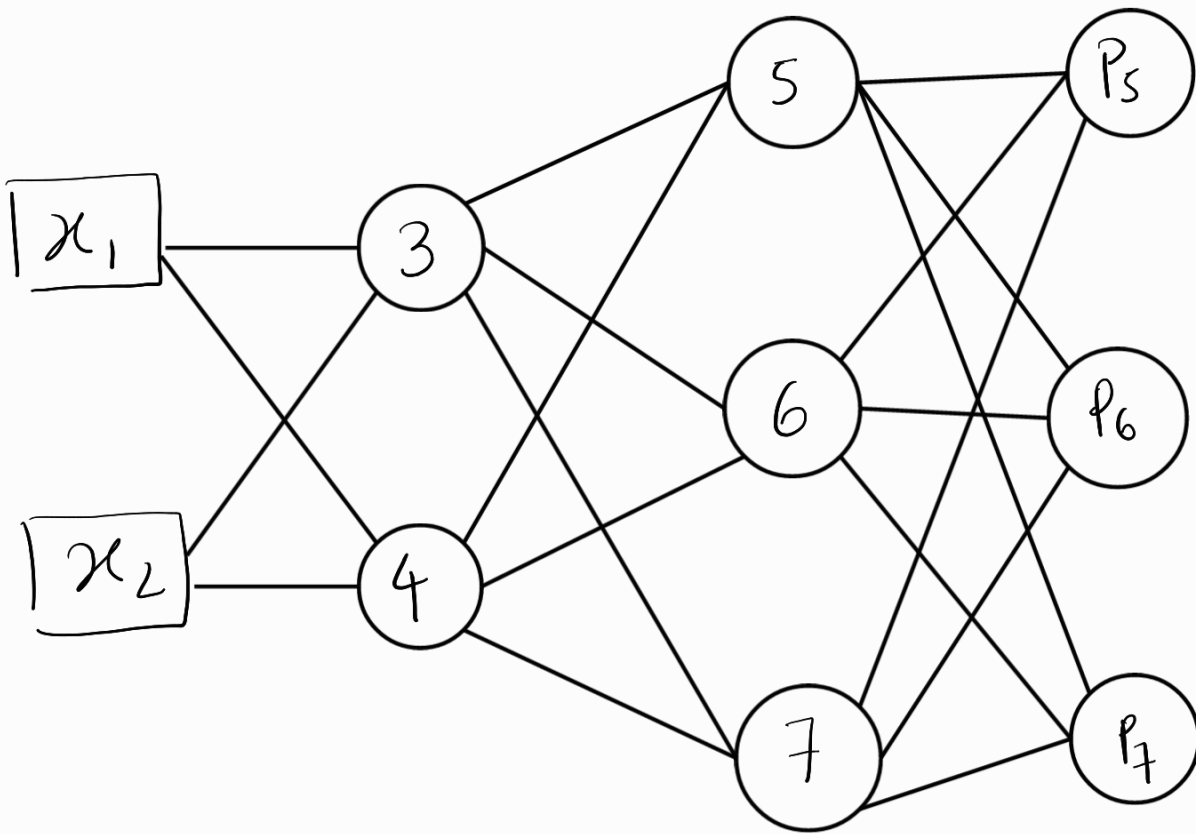


Softmax Proof



$$Y_3 = \text{ReLU}(W_{13}x_1 + W_{23}x_2 + b_3)$$

$$Y_4 = \text{ReLU}(W_{14}x_1 + W_{24}x_2 + b_4)$$

$$Y_5 = W_{35}Y_3 + W_{45}Y_4 + b_5$$

$$Y_6 = W_{36}Y_3 + W_{46}Y_4 + b_6$$

$$Y_7 = W_{37}Y_3 + W_{47}Y_4 + b_7$$

$$P_5 = \frac{e^{y_5}}{e^{y_5} + e^{y_6} + e^{y_7}}$$

$$P_6 = \frac{e^{y_6}}{e^{y_5} + e^{y_6} + e^{y_7}}$$

$$P_7 = \frac{e^{y_7}}{e^{y_5} + e^{y_6} + e^{y_7}}$$

Say, target output $y_5 = 1$

then

$$\text{Loss} = - \sum y \ln \hat{y}$$

only for node 5, $y = 1$

So,

$$\text{Loss} = - \ln P_5 - 0 - 0$$

$$= - \ln P_5$$

Now

$$\frac{d\text{Loss}}{dP_5} = - \frac{1}{P_5}$$

$$\frac{dP_5}{dY_5} = \frac{d}{dY_5} \left(\frac{e^{Y_5}}{e^{Y_5} + e^{Y_6} + e^{Y_7}} \right)$$

$$= \frac{e^{Y_5} (e^{Y_5} + e^{Y_6} + e^{Y_7}) - e^{Y_5} \cdot e^{Y_5}}{(e^{Y_5} + e^{Y_6} + e^{Y_7})^2}$$

$$= \frac{e^{Y_5} (e^{Y_5} + e^{Y_6} + e^{Y_7} - e^{Y_5})}{(e^{Y_5} + e^{Y_6} + e^{Y_7})^2}$$

$$= \left(\frac{e^{Y_5}}{e^{Y_5} + e^{Y_6} + e^{Y_7}} \right) \times \frac{e^{Y_5} + e^{Y_6} + e^{Y_7} - e^{Y_5}}{e^{Y_5} + e^{Y_6} + e^{Y_7}}$$

$$= P_5 \times \left(\frac{e^{Y_5} + e^{Y_6} + e^{Y_7}}{e^{Y_5} + e^{Y_6} + e^{Y_7}} - \frac{e^{Y_5}}{e^{Y_5} + e^{Y_6} + e^{Y_7}} \right)$$

$$= P_5 \times (1 - P_5)$$

Now,

$$\begin{aligned}\frac{d \text{Loss}}{d \gamma_5} &= \frac{d \text{Loss}}{d p_5} \times \frac{d p_5}{d \gamma_5} \\ &= -\frac{1}{p_5} \times p_5 (1 - p_5) \\ &= p_5 - 1\end{aligned}$$

Now, for other node (6 and 7)

$$\begin{aligned}\frac{d \text{Loss}}{d \gamma_6} &= \frac{d \text{Loss}}{d p_5} \times \frac{d p_5}{d \gamma_6} \\ &= -\frac{1}{p_5} \times \frac{d p_5}{d \gamma_6}\end{aligned}$$

$$\frac{d p_5}{d \gamma_6} = \frac{d}{d \gamma_6} \left(\frac{e^{\gamma_5}}{e^{\gamma_5} + e^{\gamma_6} + e^{\gamma_7}} \right)$$

$$= \frac{0 - e^{\gamma_5} e^{\gamma_6}}{(e^{\gamma_5} + e^{\gamma_6} + e^{\gamma_7})^2}$$

$$= - \frac{e^{\gamma_5}}{e^{\gamma_5} + e^{\gamma_6} + e^{\gamma_7}} \times \frac{e^{\gamma_6}}{e^{\gamma_5} + e^{\gamma_6} + e^{\gamma_7}}$$

$$= - P_5 P_6$$

$$\therefore \frac{\partial \text{Loss}}{\partial \gamma_6} = \frac{\partial \text{loss}}{\partial P_5} \times \frac{\partial P_5}{\partial \gamma_6}$$

$$= - \frac{1}{P_5} \times - P_5 P_6$$

$$= P_6$$

Same for 7th node